

GATE EE Batch 004

Human Final QA - Mobile Fillable Review Packet

Human reviewer record

Name:

Role or qualification:

Review date (YYYY-MM-DD):

Reviewer identifier (optional):

Type the attestation exactly as printed below:

This packet contains 15 checksum-bound Batch 004 Signals and Systems questions. Mathematical expressions are canonical LaTeX; declared SVG diagrams are checksum-bound and rendered below their stems.

Canonical source SHA-256: 0d0eb24d9f82f1f511f8dcb879dd78d5684fa73c444a8319d6bc01bdd7961b66

Strict Formatter report SHA-256: 25f128b7778a9504e0ddd76380b91be92fea82511952f400ede72f827b201834

Automated prerequisite: **15 PASS / 0 REVIEW / 0 invalid**. This does not replace your human judgment.

Required procedure

1. Solve each question independently before relying on the declared answer.
2. Inspect every displayed diagram and all labels when a figure is present.
3. Select exactly one of PASS or FAIL for each of the five checks.
4. Select exactly one final decision: PASS, REVISE, or REJECT.
5. Explain every REVISE or REJECT decision in Notes.
6. Save this filled PDF and return it for checksum-bound import and validation.

Required attestation: I independently reviewed the Batch 004 questions, answers and solutions and approve only the question IDs marked PASS below.

State your actual background accurately in the qualification field; do not invent a title or credential.

TMB-GATE-EE-SS-001 - revision 1

Question 1 of 15

Route: Electrical Engineering -> Signals and Systems -> Continuous and Discrete Signals

Type / marks / difficulty: MCQ / 1 / Easy

Question

The continuous-time signal $x(t) = \cos(4t) + \sin(6t)$ is periodic. Determine its fundamental period.

Options

- A. $\frac{\pi}{3}$ s
- B. $\frac{\pi}{2}$ s
- C. π s
- D. 2π s

Verification reference

Declared answer: C

Independent recomputation evidence: The component angular frequencies have greatest common divisor 2 rad/s, giving $T_0 = \pi$ s.

Canonical solution:

The angular frequencies are 4 and 6 rad/s. Their greatest common divisor is $\omega_0 = 2$ rad/s, so the fundamental period is $T_0 = \frac{2\pi}{\omega_0} = \pi$ s. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-002 - revision 1

Question 2 of 15

Route: Electrical Engineering -> Signals and Systems -> Continuous and Discrete Signals

Type / marks / difficulty: NAT / 1 / Easy

Question

For the discrete-time signal $x[n] = (\frac{1}{2})^n u[n]$, determine the signal energy $\sum_{n=-\infty}^{\infty} |x[n]|^2$.

Verification reference

Declared answer: 1.333

Independent recomputation evidence: The squared sequence is $(1/4)^n u[n]$, whose geometric sum is $4/3 = 1.333$.

Canonical solution:

Because $u[n]$ restricts the sequence to $n \geq 0$, the energy is $E = \sum_{n=0}^{\infty} (\frac{1}{4})^n = \frac{1}{1-1/4} = \frac{4}{3} = 1.333$ to three decimal places. Final answer: 1.333

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-003 - revision 1

Question 3 of 15

Route: Electrical Engineering -> Signals and Systems -> Continuous and Discrete Signals

Type / marks / difficulty: MSQ / 2 / Hard

Question

A continuous-time system is defined by $y(t) = tx(t)$. Select all properties that this system satisfies.

Options

- A. Linear
- B. Time invariant
- C. Memoryless
- D. Causal

Verification reference

Declared answer: A, C, D

Independent recomputation evidence: Multiplication by t is linear, memoryless and causal, but a time shift does not commute with the multiplier.

Canonical solution:

Superposition is preserved because multiplication by the fixed function t is linear. The output at t depends only on $x(t)$, so the system is memoryless and causal. For a shifted input, the response is $tx(t - t_0)$, whereas a shifted version of the original output is $(t - t_0)x(t - t_0)$; hence the system is time varying. Final answer: A, C, D

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-004 - revision 1

Question 4 of 15

Route: Electrical Engineering -> Signals and Systems -> Continuous and Discrete Signals

Type / marks / difficulty: MCQ / 1 / Medium

Question

The triangular signal shown in the figure is $x(t) = 2 - |t|$ for $|t| \leq 2$ and zero otherwise. Determine its energy.

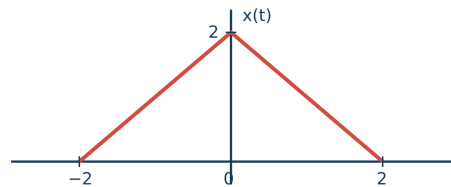


Figure: Triangular signal $x(t)$

Diagram description: Triangular signal with vertices at $(-2,0)$, $(0,2)$, and $(2,0)$.

Options

- A. $\frac{8}{3}$
- B. $\frac{4}{3}$
- C. $\frac{16}{3}$
- D. 8

Verification reference

Declared answer: C

Independent recomputation evidence: Even symmetry gives $2 \int_0^2 (2-t)^2 dt = 16/3$.

Canonical solution:

Using even symmetry, $E = 2 \int_0^2 (2-t)^2 dt = 2[-(2-t)^3/3]_0^2 = \frac{16}{3}$. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-005 - revision 1

Question 5 of 15

Route: Electrical Engineering -> Signals and Systems -> LTI Systems and Convolution

Type / marks / difficulty: NAT / 2 / Medium

Question

Let $x(t) = e^{-t}u(t)$ and $h(t) = e^{-2t}u(t)$. If $y(t) = x(t) * h(t)$, determine $y(\ln 2)$.

Verification reference

Declared answer: 0.25

Independent recomputation evidence: Direct convolution gives $e^{-t} - e^{-2t}$ for $t \geq 0$ and hence $1/4$ at $t = \ln 2$.

Canonical solution:

For $t \geq 0$, $y(t) = \int_0^t e^{-\tau} e^{-2(t-\tau)} d\tau = e^{-2t}(e^t - 1) = e^{-t} - e^{-2t}$. At $t = \ln 2$, this is $\frac{1}{2} - \frac{1}{4} = 0.25$.
Final answer: 0.25

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-006 - revision 1

Question 6 of 15

Route: Electrical Engineering -> Signals and Systems -> LTI Systems and Convolution

Type / marks / difficulty: NAT / 1 / Easy

Question

A real LTI system has frequency response $H(j2) = 3e^{j\pi/6}$. For input $x(t) = \cos(2t + \pi/18)$, enter the amplitude of the steady-state output sinusoid.

Verification reference

Declared answer: 3

Independent recomputation evidence: The LTI frequency-response magnitude at 2 rad/s is 3, so the output amplitude is 3.

Canonical solution:

A sinusoid is an eigenfunction of an LTI system. Its amplitude is multiplied by $|H(j2)| = 3$, while its phase is shifted by $\pi/6$. Therefore the output amplitude is 3. Final answer: 3

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-007 - revision 1

Question 7 of 15

Route: Electrical Engineering -> Signals and Systems -> LTI Systems and Convolution

Type / marks / difficulty: MCQ / 2 / Medium

Question

For a discrete-time LTI system, $x[n] = h[n] = u[n] - u[n - 3]$. Determine the output sample $y[2]$.

Options

- A. 1
- B. 2
- C. 3
- D. 4

Verification reference

Declared answer: C

Independent recomputation evidence: At $n = 2$, three overlapping unit samples contribute to the convolution.

Canonical solution:

Both $x[n]$ and $h[n]$ equal one at $n = 0, 1, 2$. Hence $y[2] = \sum_k x[k]h[2 - k]$ has contributions for $k = 0, 1, 2$, each equal to one. Thus $y[2] = 3$. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-008 - revision 1

Question 8 of 15

Route: Electrical Engineering -> Signals and Systems -> Fourier Analysis

Type / marks / difficulty: MSQ / 1 / Medium

Question

The periodic waveform shown in the figure equals $+2$ for $0 \leq t < T/4$ and -2 for $T/4 \leq t < T$. Select all correct statements.

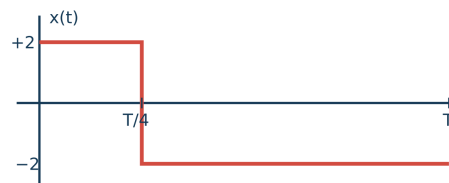


Figure: Two-level periodic waveform

Diagram description: One period of a waveform at $+2$ for the first quarter-period and -2 for the remaining three quarters.

Options

- A. Its average value is -1 .
- B. Its RMS value is 2 .
- C. Its average power is 4 .
- D. Its DC component is zero.

Verification reference

Declared answer: A, B, C

Independent recomputation evidence: Weighted averaging gives -1 ; constant squared magnitude gives RMS 2 and power 4 .

Canonical solution:

The average is $2(1/4) - 2(3/4) = -1$. Since $x^2(t) = 4$ throughout the period, the RMS value is $\sqrt{4} = 2$ and the average power is 4 . The nonzero average means the DC component is not zero. Final answer: A, B, C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-009 - revision 1

Question 9 of 15

Route: Electrical Engineering -> Signals and Systems -> Fourier Analysis

Type / marks / difficulty: NAT / 2 / Hard

Question

The periodic input to a real LTI system is $x(t) = 2 + 4 \cos(\omega_0 t) + 3 \sin(2\omega_0 t)$. The frequency-response magnitudes are $|H(0)| = \frac{1}{2}$, $|H(j\omega_0)| = 1$, and $|H(j2\omega_0)| = 2$. Determine the average output power.

Verification reference

Declared answer: 27

Independent recomputation evidence: Orthogonal output components contribute $1^2 + 4^2/2 + 6^2/2 = 27$.

Canonical solution:

The output DC value is $2(1/2) = 1$. The two sinusoidal output amplitudes are 4 and 6. Orthogonality of distinct harmonics gives $P_y = 1^2 + \frac{4^2}{2} + \frac{6^2}{2} = 1 + 8 + 18 = 27$. Final answer: 27

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-010 - revision 1

Question 10 of 15

Route: Electrical Engineering -> Signals and Systems -> Laplace Transform

Type / marks / difficulty: MCQ / 1 / Easy

Question

The signal $x(t) = e^{-2t}u(t) - e^{3t}u(-t)$ has a bilateral Laplace transform. Which region of convergence applies?

Options

- A. $\text{Re}(s) > 3$
- B. $\text{Re}(s) < -2$
- C. $-2 < \text{Re}(s) < 3$
- D. $\text{Re}(s) > -2$

Verification reference

Declared answer: C

Independent recomputation evidence: The right-sided term requires $\text{Re}(s) > -2$ and the left-sided term requires $\text{Re}(s) < 3$.

Canonical solution:

The right-sided term requires $\text{Re}(s) > -2$. The left-sided term requires $\text{Re}(s) < 3$. Their intersection is $-2 < \text{Re}(s) < 3$. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-011 - revision 1

Question 11 of 15

Route: Electrical Engineering -> Signals and Systems -> Laplace Transform

Type / marks / difficulty: NAT / 2 / Medium

Question

A causal signal has Laplace transform $X(s) = \frac{s+4}{s(s+2)(s+3)}$. Determine its final value $\lim_{t \rightarrow \infty} x(t)$.

Verification reference

Declared answer: 0.667

Independent recomputation evidence: The final-value theorem gives $\lim_{s \rightarrow 0} (s+4)/[(s+2)(s+3)] = 2/3 = 0.667$.

Canonical solution:

All poles of $sX(s)$ lie in the open left half-plane, so the final-value theorem applies. Hence $x(\infty) = \lim_{s \rightarrow 0} sX(s) = \frac{4}{(2)(3)} = \frac{2}{3} = 0.667$ to three decimal places. Final answer: 0.667

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-012 - revision 1

Question 12 of 15

Route: Electrical Engineering -> Signals and Systems -> Z Transform

Type / marks / difficulty: MCQ / 1 / Medium

Question

For $X(z) = \frac{z}{z-0.5} - \frac{z}{z-2}$ with region of convergence $|z| > 2$, determine $x[n]$.

Options

- A. $[(0.5)^n - 2^n]u[n]$
- B. $[(0.5)^n - 2^n]u[-n - 1]$
- C. $(0.5)^n u[n] + 2^n u[-n - 1]$
- D. $-(0.5)^n u[-n - 1] - 2^n u[n]$

Verification reference

Declared answer: A

Independent recomputation evidence: The ROC outside both poles makes both inverse-Z components right sided.

Canonical solution:

The stated region lies outside both poles, so both components are right sided. Using $\frac{z}{z-a} \leftrightarrow a^n u[n]$ gives $x[n] = (0.5)^n u[n] - 2^n u[n]$. Final answer: A

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-013 - revision 1

Question 13 of 15

Route: Electrical Engineering -> Signals and Systems -> Z Transform

Type / marks / difficulty: MSQ / 2 / Hard

Question

Consider $H(z) = \frac{1-0.5z^{-1}}{(1-0.25z^{-1})(1-2z^{-1})}$. Select all correct statements about possible regions of convergence.

Options

- A. For $|z| > 2$, the system is causal but unstable.
- B. For $0.25 < |z| < 2$, the system is stable and two sided.
- C. For $|z| < 0.25$, the system is anti-causal and unstable.
- D. The system is FIR for every possible region of convergence.

Verification reference

Declared answer: A, B, C

Independent recomputation evidence: Only the annular ROC contains the unit circle; outer and inner ROCs are causal and anti-causal respectively and both unstable.

Canonical solution:

An outer ROC gives a right-sided causal impulse response, but it excludes the unit circle and is unstable. The annular ROC produces a two-sided response and includes the unit circle, so it is stable. The inner ROC gives a left-sided anti-causal response and excludes the unit circle, so it is unstable. The two uncanceled poles make the response IIR. Final answer: A, B, C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-014 - revision 1

Question 14 of 15

Route: Electrical Engineering -> Signals and Systems -> Sampling

Type / marks / difficulty: NAT / 1 / Medium

Question

The magnitude spectrum shown in the figure is nonzero only over $2 \leq |f| \leq 4$ kHz. Using the conventional baseband sampling theorem, enter the minimum sampling frequency in kHz that guarantees no aliasing.

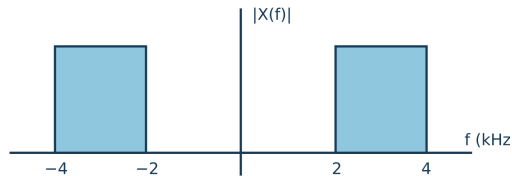


Figure: Magnitude spectrum $|X(f)|$

Diagram description: Two rectangular spectral bands from -4 to -2 kHz and from 2 to 4 kHz.

Verification reference

Declared answer: 8

Independent recomputation evidence: The highest occupied frequency is 4 kHz, so conventional Nyquist sampling requires 8 kHz.

Canonical solution:

The highest frequency present is 4 kHz. The conventional Nyquist criterion requires $f_s \geq 2f_{\max} = 8$ kHz.
Final answer: 8

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-SS-015 - revision 1

Question 15 of 15

Route: Electrical Engineering -> Signals and Systems -> Sampling

Type / marks / difficulty: MCQ / 2 / Hard

Question

The signal $x(t) = \cos(2\pi 1000t) + \cos(2\pi 2600t)$ is sampled at 4 kHz. The samples are passed through an ideal reconstruction low-pass filter with passband $0 \leq f \leq 2$ kHz. Which two positive-frequency components appear in the reconstructed signal?

Options

- A. 0.6 kHz and 1.0 kHz
- B. 1.0 kHz and 1.4 kHz
- C. 1.0 kHz and 2.0 kHz
- D. 1.4 kHz and 2.6 kHz

Verification reference

Declared answer: B

Independent recomputation evidence: Sampling at 4 kHz folds 2.6 kHz to 1.4 kHz while preserving 1 kHz.

Canonical solution:

The 1 kHz component lies below Nyquist and is unchanged. The 2.6 kHz component folds about 4 kHz to $|2.6 - 4| = 1.4$ kHz. Both 1.0 and 1.4 kHz pass through the reconstruction filter. Final answer: B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

Review completion

Complete this page only after reviewing all 15 questions.

Final reviewed-results decision: APPROVE REVIEWED RESULTS ☐

Reviewer typed signature:

Final comments:

Paper eligibility remains blocked until this completed record is validated and only explicitly approved IDs are promoted.